

MSDM 5056 Network Modeling
Assignment 4 (due 19th November, 2025)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Building a small-world network

A small-world network can be built by “rewiring” edges of a c -regular ring lattice (refer to p.23–24 in Lecture 7), but the precise details of the rewiring process are up to our interpretation. For example, we may rewire an edge by first removing it and then adding an edge that joins any two vertices of the whole network (i.e. including the two ends of the original edge). Although this may result in multiple edges between two vertices, this problem is insignificant once the network size $n \gg c$.

- (a) A more serious problem of such a rewiring process is that a vertex can become disconnected from the rest of the network if it receives no new edges after having all of its c edges removed. Show that the probability of this happening to any given vertex is $(pe^{-p})^c$, where p is the rewiring probability.
- (b) The current world population is 8 billion. If all people of this world form a small-world network with $c = 8$ through such a rewiring process, at what value of p would you expect one person to be isolated from the rest of the world?

Solution:

- (a) The probability that a specific vertex has all of its c edges removed for rewiring reads p^c . The probability of adding an edge which connects to the specific vertex reads $\frac{n-1}{n(n-1)/2} = \frac{2}{n}$, so the probability of adding no such edges reads $\left(1 - \frac{2}{n}\right)^{\frac{ncp}{2}}$, where $\frac{nc}{2}$ is the number of edges of the lattice and $\frac{ncp}{2}$ is thus the number of rewired edges. In the large n limit, $\left(1 - \frac{2}{n}\right)^{\frac{ncp}{2}}$ becomes e^{-cp} . Hence, the overall probability is $p^c e^{-cp} = (pe^{-p})^c$.
- (b) Using the result from (a), one has $n(pe^{-p})^c = 1$ or $(pe^{-p})^8 = \frac{1}{8 \times 10^9}$, so $p \approx 0.0616$.

(2) Network evolution

An undirected network is growing in the following way. At time $t = 1$, it consists of a single isolated node. At every time step, a new node is added to the network and then connected to an existing node by a new link.

- (a) What is the total number of nodes N at time t ?
- (b) What is the total number of links L at time t ?
- (c) What is the average degree $\langle k \rangle$ at time t ?
- (d) What is the average degree $\langle k \rangle$ when $t \rightarrow \infty$?

Solution:

- (a) $N = t$
- (b) $L = t - 1$
- (c) $\langle k \rangle = \frac{2L}{N} = \frac{2(t-1)}{t} = 2 \left(1 - \frac{1}{t}\right)$
- (d) When $t \rightarrow \infty$, we have $\lim_{T \rightarrow \infty} \langle k \rangle = \lim_{T \rightarrow \infty} 2 \left(1 - \frac{1}{t}\right) = 2$.

(3) The Barabasi-Albert model

The Barabasi-Albert model is a growing network model that exhibits a power-law degree distribution. It can be built in the following way. At time $t = 1$, the network is formed by two nodes joined by a link. At every time step, a single new node joins the network so that at time t there will be exactly $N = 1 + t$ nodes. Every new node has initially $m = 1$ link. Each new link is attached to an existing node i of the network, which is chosen with probability P_i according to the preferential attachment rule, i.e.

$$P_i = \frac{k_i}{\sum_{j=1}^n k_j}$$

where k_i is the degree of i . In order to calculate properties of the model, mean-field approximation assumes that degree is a continuous number.

- (a) What is the time evolution of $k_i = k_i(t)$ in the mean-field approximation? (*Refer to p.42 or Lecture 8*)
- (b) What is the maximum degree in the network predicted by the mean-field approximation?
- (c) What is the degree distribution of the network in the mean-field approximation after a long time? What would be the behavior of the cumulative distribution function of this degree distribution?
- (d) Use the result in (c) to calculate $\langle k^2 \rangle$ in the mean-field approximation, where $\langle \dots \rangle$ indicates an average over the degree distribution of the network.

- (e) Is the Barabasi-Albert model a scale-free network? Why?
- (f) One can indeed obtain the exact degree distribution $P(k)$ of the network by solving a master equation of the expected number of nodes $n_k(t)$ that have degree k at time t . It takes the form

$$n_k(t+1) = n_k(t) + \frac{k-1}{2t}n_{k-1}(t) - \frac{k}{2t}n_k(t)$$

for $k > 1$ and

$$n_1(t+1) = n_1(t) - \frac{1}{2t}n_1(t) + 1$$

for $k = 1$. (Convince yourself that this master equation describes the change of n_k as time evolves.)

- (i) Assume $n_k(t) = tP(k)$ as $t \gg 1$. By substituting $P(k)$ into the master equation, show that

$$P(k) = \frac{k-1}{k+2}P(k-1)$$

- (ii) Show that the recursion relation in (i) has the following solution

$$P(k) = \frac{C}{k(k+1)(k+2)}$$

where C is a constant.

- (iii) What is the value of C ? (Hint: Use what you obtain for $P(1)$ to get C)
- (iv) Compare the result in (c) with the result you obtain here.

Solution:

- (a) In the mean-field approximation, $k_i(t)$ follows the following differential equation

$$\frac{d}{dt}k_i(t) = m \frac{k_i(t)}{\sum_j k_j(t)}$$

Here $k_i(t_i) = m = 1$, where t_i is the arrival time of i in the network. The number of links at time t is $L(t) = t$. Therefore, $\sum_j k_j(t) = 2L(t) = 2t$. Solving the differential equation gives

$$k_i(t) = \left(\frac{t}{t_i}\right)^{1/2}$$

- (b) The maximum degree predicted by the mean-field approximation is $k_i(t) = \sqrt{t}$, which is the degree of the node arrived in the network at time $t_i = 1$
- (c) The degree distribution in the mean-field approximation after a long time is given by $P(k) \sim \frac{2}{k^3}$. Let its cumulative distribution function be $P_{CDF}(k)$.

$$P_{CDF}(k) = \int_{k_{min}}^k P(k')dk' = \int_1^k \frac{2}{(k')^3}dk' \sim \frac{1}{k^2}$$

Therefore, the tail of $P_{CDF}(k)$ also follows a power law.

(d) In the mean-field approximation, the degree distribution from (c) gives

$$\begin{aligned}\langle k^2 \rangle &= \int_1^{\sqrt{t}} k^2 P(k) dk \\ &\sim \int_1^{\sqrt{t}} k^2 \frac{2}{k^3} dk = 2 \ln(\sqrt{t}) = \ln t\end{aligned}$$

Note that $\langle k^2 \rangle \rightarrow \infty$ as $t \rightarrow \infty$. In other words, the Barabasi-Albert model gives an infinite variance of degree.

(e) The Barabasi-Albert model is a scale-free network since its degree distribution $P(k) \sim k^{-3}$ follows a power law.

(f)

(i) Substituting $n_k(t) = tP(k)$ into the master equation, we get

$$P(k) = \frac{k-1}{2} P(k-1) - \frac{k}{2} P(k)$$

for $k > 1$ and $t \gg 1$, which is equivalent to

$$P(k) = \frac{k-1}{k+2} P(k-1)$$

(ii) One can show this by direct substitution. If the solution is true, the LHS of the recursion relation is $\frac{C}{k(k+1)(k+2)}$, and the RHS is

$$\frac{k-1}{k+2} \frac{C}{(k-1)k(k+1)} = \frac{C}{k(k+1)(k+2)}$$

so the solution satisfies the recursion relation.

(iii) Substituting $n_1(t) = tP(1)$ into the master question, we get

$$(t+1)P(1) = tP(1) - \frac{1}{2t} tP(1) + 1$$

as $k = 1$ and $t \gg 1$. This gives

$$P(1) = \frac{2}{3} = \frac{4}{1 \times (1+1) \times (1+2)}$$

and thus $C = 4$.

(iv) When k is large, both results give a power-law distribution $P(k) \sim k^{-3}$.